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Edge-Deployable Skin Lesion Classification with a Quantized CNN-Transformer Ensemble

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Abstract

Automated classification of skin lesions using dermoscopic images can assist in distinguishing visually similar lesion categories. This work evaluates MobileNetV2 and DINOv2 for classifying Basal Cell Carcinoma (BCC), Melanoma, and Nevus from a curated three-class subset of the ISIC 2019 dataset. The study focuses on a reproducible comparison of full-precision, quantized, and pruned models, together with a probability-level ensemble analysis and explainability assessment. On the existing 2,489-image image-level validation partition, MobileNetV2 FP32 achieved 90.96% accuracy and 87.06% Melanoma recall. Full-integer INT8 TFLite conversion reduced the model size from 30.21 MiB to 2.94 MiB, while retaining 90.72% accuracy and 86.73% Melanoma recall; host-CPU batch-one latency was 4.51 ms. In contrast, 50% unstructured pruning reduced accuracy to 83.49%, while native dynamic INT8 quantization of DINOv2 provided a compact 87.38 MiB representation and 173.54 ms host latency, with accuracy of 61.59% under the evaluated quantization configuration. This result identifies a clear opportunity for quantization-aware training and distillation. An equal-weight ensemble achieved 91.36% on the same validation partition, but this result is not an independent test result because lesion-level auditing identified 54.04% overlap between validation and training images. The results therefore support comparative compression and deployment analysis, but not clinical generalization. Raspberry Pi measurements and grouped held-out test evaluations remain necessary before making an on-device or clinical performance claim.

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Keywords: Skin lesion classification; Weighted ensemble; MobileNetV2; DINOv2; Edge deployment; ISIC 2019

1. Introduction

Skin lesion analysis plays an important role in the identification and assessment of dermatological conditions. BCC, Melanoma, and Nevus may exhibit visual similarities, making accurate image-based classification challenging. MobileNetV2 uses inverted residual blocks and depthwise separable convolutions to reduce computational complexity [3], whereas DINOv2 provides a transformer-based self-supervised representation [5]. Evaluating these complementary architectures is useful for understanding the accuracy–efficiency trade-off in resource-constrained applications.

This study uses a curated three-class subset of ISIC 2019. The selected classes provide a focused evaluation of BCC, Melanoma, and Nevus, with particular attention to the clinically important confusion between Melanoma and Nevus. The experiments compare FP32 models, MobileNetV2 Dynamic-Range and Full INT8 TFLite models, a 50%

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unstructured-pruned MobileNetV2, and DINOv2 dynamic INT8 diagnostics. A probability-level ensemble is also audited, but its results are interpreted as validation results rather than independent test performance.

The main contributions of this work are summarized as follows:

- Development of a three-class skin lesion classification framework for BCC, Melanoma, and Nevus.
- Evaluation of MobileNetV2 and DINOv2 using 50-epoch training experiments.
- Analysis of training and validation accuracy and loss using the recorded epoch-wise experimental histories.
- Implementation of a weighted probability-level ensemble combining MobileNetV2 and DINOv2 predictions.
- Quantitative comparison of model accuracy, class-level recall, storage size, and host-CPU latency across seven model variants.
- Analysis of the trade-off between Full INT8 quantization and 50% unstructured pruning for MobileNetV2.
- Diagnosis of DINOv2 activation outliers that define the optimization target for quantization-aware training and distillation.
- Explicit auditing of lesion overlap and transparent separation of comparative validation evidence from independent clinical test evidence.
- Integration of the MobileNetV2 inference pipeline into a Streamlit-based application, while identifying Raspberry Pi measurement as future work.

2. Related Work

2.1. Convolutional Approaches to Skin Lesion Classification

Deep convolutional neural networks have demonstrated strong performance in automated skin lesion analysis. Esteva et al. demonstrated dermatologist-level skin cancer classification using deep neural networks, highlighting the potential of deep learning for dermatological image analysis [2].

Convolutional neural networks such as ResNet and MobileNet have subsequently been applied to image classification problems because of their ability to learn hierarchical visual features [8, 4]. MobileNetV2 improves computational efficiency using inverted residual structures and linear bottlenecks [3], making it suitable for applications requiring lightweight models.

2.2. Transformer-Based Visual Representation Learning

Vision Transformers provide an alternative approach to image analysis by modelling relationships between image regions using attention mechanisms [7]. Self-supervised learning approaches further improve the ability of models to learn transferable visual representations without requiring large quantities of labelled data.DINO introduced self-distillation for learning visual representations [6], while DINOv2 extended this approach to produce robust and transferable visual features suitable for multiple downstream computer vision tasks [5]. Such representations can be adapted for medical image classification through fine-tuning and task-specific classification layers.

2.3. Ensemble Learning

Ensemble learning combines predictions generated by multiple classifiers to improve overall performance. The potential benefit of an ensemble depends on the diversity of the individual models and the differences in their prediction errors [13, 14].Combining convolutional and transformer-based architectures may provide complementary information because the two models learn visual representations using different architectural mechanisms. This work investigates weighted probability-level fusion of MobileNetV2 and DINOv2 for three-class skin lesion classification.

3. Proposed Methodology

Figure 1 presents the overall architecture of the proposed skin lesion classification framework. The input dermoscopic image undergoes preprocessing before being provided independently to MobileNetV2 and DINOv2. The probability predictions generated by both models are combined using a weighted ensemble strategy. The final class is selected based on the highest ensemble probability and presented through the deployed application.

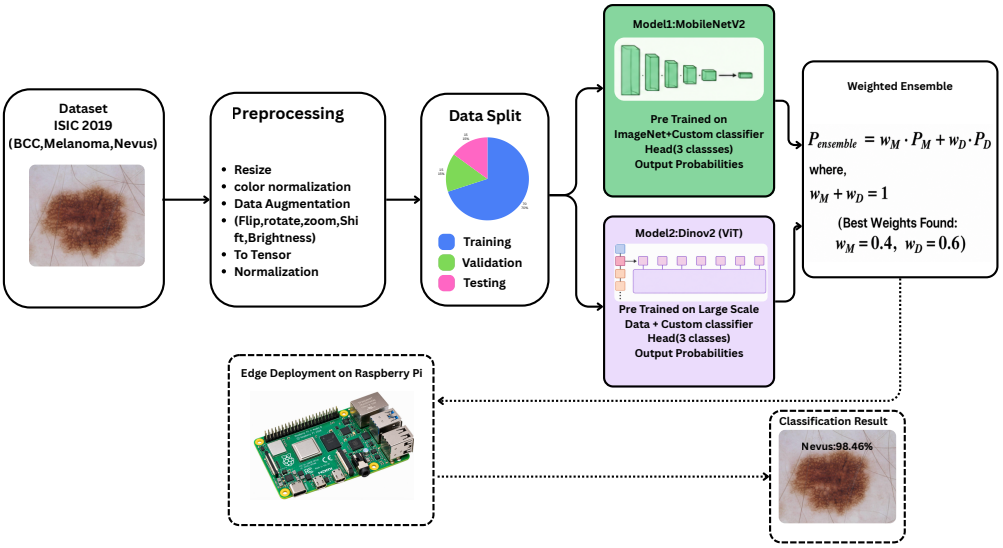


Fig. 1. Overall architecture of the proposed edge-deployable skin lesion classification framework.

3.1. Dataset

The experimental study was conducted using a selected three-class subset of the ISIC 2019 skin lesion dataset [1]. The three categories considered in this work were Basal Cell Carcinoma (BCC), Melanoma, and Nevus.These categories include malignant and benign skin lesion types and establish a multi-class classification task. The class labels were maintained consistently throughout the experiments to ensure correct alignment between the ground-truth labels and the prediction probabilities generated by MobileNetV2 and DINOv2.Representative input images from the three selected categories are shown in Figure 7. The images demonstrate examples of BCC, Melanoma, and Nevus lesions used as inputs for the classification framework.

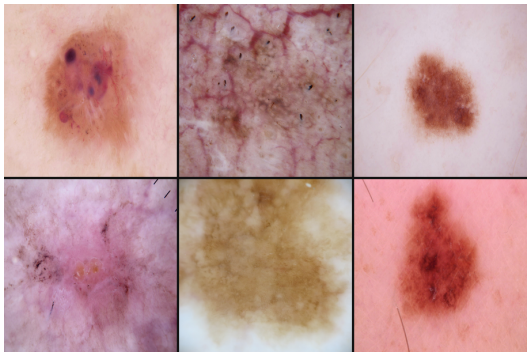


Fig. 2. Representative dermoscopic input images from the dataset

The class mapping used in this work is shown in Table 1.

Table 1. Class mapping used for skin lesion classification.

Class Label	Skin Lesion Category
0	BCC
1	Melanoma
2	Nevus

For final comparative evaluation and ensemble analysis, an image-level validation partition containing 2,489 images was used: 665 BCC, 904 Melanoma, and 920 Nevus images. The same class-label mapping and image ordering were maintained while generating predictions from MobileNetV2 and DINOv2 so that the corresponding probability vectors could be combined correctly. This partition is not treated as an independent test set.

3.2. Data Splitting and Leakage Audit

The legacy evaluation protocol used a stratified image-level 80:20 split, producing 9,956 training images and 2,489 validation images. A subsequent lesion-ID audit identified 5,502 unique known lesions and 1,886 images without a lesion ID. Of the validation images, 1,345 (54.04%) belonged to lesions also represented in the training partition. The overlap was particularly high for BCC (85.11% of validation images) and Melanoma (79.31%), while the corresponding Nevus overlap was 6.74%.

The overlap materially changes the interpretation of the results. On the clean non-overlapping subset of 1,144 validation images, MobileNetV2 accuracy decreased from 90.96% to 86.80%, DINOv2 accuracy decreased from 86.66% to 84.00%, and the equal-weight ensemble decreased from 91.36% to 87.15%. Lesion-grouped Revision V1 split manifests have been prepared, but valid retrained models and held-out test predictions are not yet available. Accordingly, the reported results quantify relative model and compression behaviour on the legacy validation partition.

3.3. Data Preprocessing

The input dermoscopic images were resized to 224×224 pixels. MobileNetV2 inputs were converted to RGB and scaled by $1/255$ to the range $[0, 1]$. DINOv2 inputs were normalized using the ImageNet mean and standard deviation. The verified MobileNetV2 training pipeline used rotation, zoom, and translation augmentation. Horizontal flipping and brightness augmentation are not claimed for the reported run. After preprocessing, the images were converted into the required tensor representation and supplied to the respective models.

3.4. MobileNetV2-Based Classification

MobileNetV2 was employed as the convolutional neural network for the skin lesion classification task. The architecture is designed to provide computational efficiency through inverted residual blocks and depthwise separable convolutions [3].The pretrained MobileNetV2 architecture was adapted for classification of the three target categories: BCC, Melanoma, and Nevus. The model was trained for 50 epochs.During training, training accuracy, validation accuracy, training loss, and validation loss were recorded for every epoch. The recorded 50-epoch history was subsequently used to generate the corresponding performance curves.The recorded epoch history reached 90.12% validation accuracy at epoch 50; the saved FP32 checkpoint achieved 90.96% on the consolidated 2,489-image evaluation partition.

3.5. DINOv2-Based Classification

DINOv2 was employed as the transformer-based model for the skin lesion classification task. The pretrained model was adapted to classify the three target categories: BCC, Melanoma, and Nevus.The DINOv2 classifier was trained for 50 epochs. Training accuracy, validation accuracy, training loss, and validation loss were recorded throughout the training process.The recorded epoch history reached 86.82% validation accuracy at epoch 35; the saved FP32 checkpoint achieved 86.66% on the consolidated evaluation partition.

3.6. Weighted Ensemble Strategy

The predictions obtained from MobileNetV2 and DINOv2 were combined using a weighted probability-level ensemble. Instead of using the prediction from a single model, the probability outputs of both models were combined to obtain the final prediction.

Let P_M represent the probability vector produced by MobileNetV2 and P_D represent the probability vector produced by DINOv2. The ensemble probability is calculated using Equation (1):

$$P_{\text{ensemble}} = w_M P_M + w_D P_D \tag{1}$$

Here, w_M and w_D represent the weights assigned to MobileNetV2 and DINOv2, respectively. The weights used in the ensemble satisfy the condition given in Equation (2):

$$w_M + w_D = 1 \tag{2}$$

The final predicted class is obtained by selecting the class with the highest probability from the ensemble output, as defined in Equation (3):

$$\hat{y} = \arg\max(P_{\text{ensemble}}) \tag{3}$$

Different combinations of weights were tested on the legacy validation partition. The audited equal-weight configuration ($w_M = 0.5$, $w_D = 0.5$) achieved 91.36% accuracy, whereas the previously reported configuration ($w_M = 0.4$, $w_D = 0.6$) achieved 90.76%. Because the weights were selected using the same validation predictions, neither result is presented as an unbiased held-out test estimate.

3.7. System Implementation and Edge Deployment

The developed skin lesion classification framework was integrated into a Streamlit-based application. The application accepts a dermoscopic skin lesion image as input and performs preprocessing before the image is provided to the trained classification pipeline.The application workflow consists of three main stages: image quality assessment, model prediction, and presentation of the classification result. After an image is uploaded, the system first performs an image quality assessment. The accepted image is then processed by the classification models, and the weighted ensemble is used to generate the final prediction.The output interface displays the predicted skin lesion category together with the associated confidence scores. This enables the user to observe the relative confidence assigned to the three classes: BCC, Melanoma, and Nevus.Figure **??** illustrates the implemented application workflow, including the diagnostic interface, image quality assessment stage, and final prediction output.

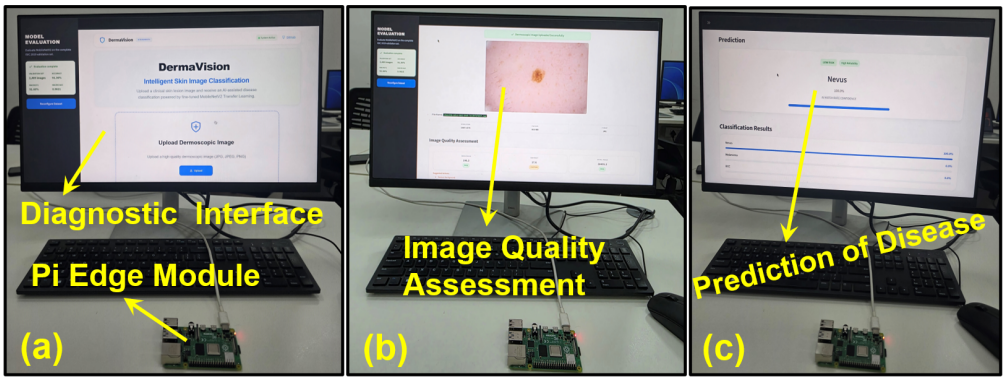


Fig. 3. Application workflow showing (a) the diagnostic interface and Pi Edge Module, (b) the image quality assessment stage, and (c) the final disease prediction with confidence scores.

The application currently uses the Dynamic-Range TFLite MobileNetV2 model. A separate Full INT8 TFLite inference module is available and is the recommended practical edge candidate based on host-CPU measurements. No Raspberry Pi benchmark result is reported in this study; therefore, Raspberry Pi latency, memory, and thermal claims are reserved for future work.

4. Experimental Results and Analysis

4.1. Experimental Setup

Both MobileNetV2 and DINOv2 were trained for 50 epochs on the selected three-class skin lesion dataset. During training, training accuracy, validation accuracy, training loss, and validation loss were recorded for every epoch. The complete recorded histories were used to generate the performance curves. The consolidated evaluation compared seven variants using accuracy, precision, recall, F1-score, model size, and batch-one host-CPU latency. The benchmark host was a Windows 11 system with a 12th Generation Intel Core i5-1235U processor and 15.65 GiB RAM. These measurements are not Raspberry Pi measurements.

4.2. Individual Model Performance

MobileNetV2 was trained for 50 epochs, and its training and validation performance was monitored throughout the training process. The recorded curves show the progression of both accuracy and loss across the epochs. The epoch history reached 90.12% at epoch 50, while the saved FP32 checkpoint achieved 90.96% on the consolidated validation partition. Figure 4 presents the recorded curves for MobileNetV2.

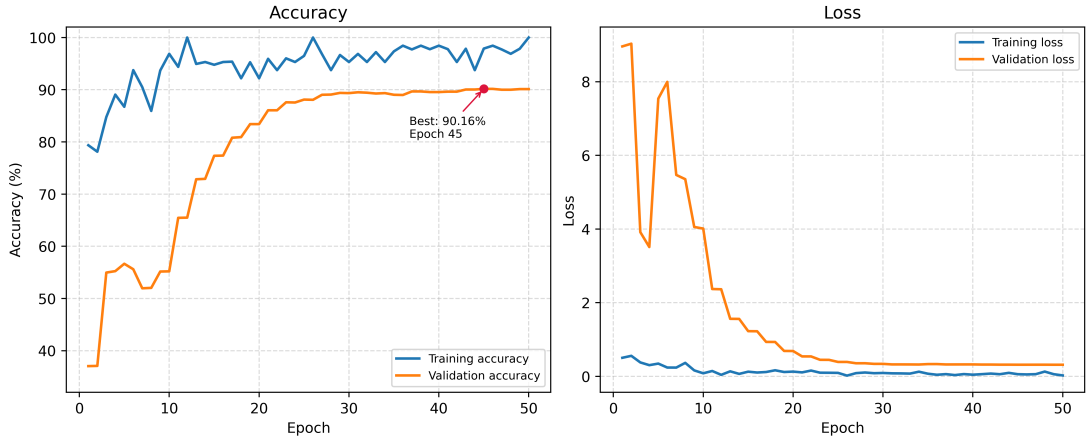


Fig. 4. Recorded training and validation accuracy and loss of MobileNetV2 over 50 epochs. The highest recorded validation accuracy was 90.16% at epoch 45; the epoch-50 value was 90.12%.

DINOv2 was also trained for 50 epochs, and its training and validation performance was observed across the training process. The epoch history reached its highest value of 86.82% at epoch 35, while the saved FP32 checkpoint achieved 86.66% on the consolidated validation partition. Figure 5 shows the recorded curves for DINOv2.

4.3. Comparison of Individual Models

The consolidated results show that MobileNetV2 FP32 achieved 90.96% accuracy, 91.27% Macro F1, and 87.06% Melanoma recall. DINOv2 FP32 achieved 86.66% accuracy, 86.82% Macro F1, and 79.09% Melanoma recall. These values are validation results from the image-level partition described in Section 3.2.

These results are summarized in Table 2.

Table 2. Recorded performance of the individual classification models.

Model	Metric	Result
MobileNetV2	Validation accuracy at epoch 50	90.12%
DINOv2	Best validation accuracy	86.82%
DINOv2	Epoch of best validation accuracy	35

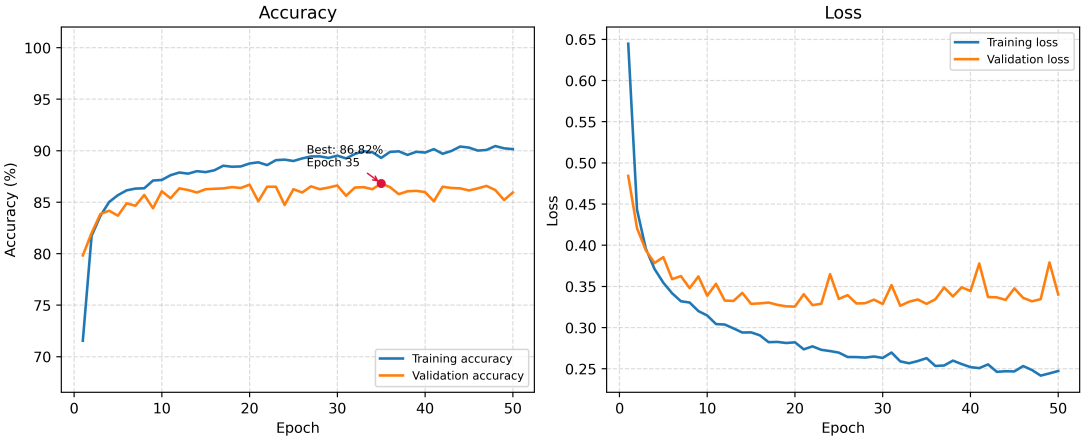


Fig. 5. Recorded training and validation accuracy and loss of DINOv2 over 50 epochs. The highest recorded validation accuracy of 86.82% was obtained at epoch 35.

Although MobileNetV2 achieved a higher recorded validation accuracy, DINOv2 provided an additional transformer-based representation. Therefore, the prediction probabilities generated by both architectures were combined using the weighted probability-level ensemble strategy.

4.4. Weighted Ensemble Performance

The prediction probabilities generated by MobileNetV2 and DINOv2 were combined using different weight configurations. The objective of this experiment was to determine whether the complementary information learned by the convolutional and transformer-based models could improve the final classification performance.The audited equal-weight ensemble ($w_M = 0.5$, $w_D = 0.5$) achieved 91.36% accuracy and 91.67% Macro F1 on the legacy validation partition. The 0.4/0.6 configuration achieved 90.76% accuracy and 91.01% Macro F1. Thus, the 91.36% result must not be attributed to the 0.4/0.6 configuration. The comparison between the standalone model and the two audited configurations is presented in Table 3.

Table 3. Comparison of MobileNetV2 and the best weighted ensemble.

Model / Configuration	**Accuracy**	**Macro F1**	**Melanoma Recall**
MobileNetV2 FP32 checkpoint	90.96%	91.27%	87.06%
Equal ensemble ($w_M = 0.5$, $w_D = 0.5$)	**91.36%**	**91.67%**	87.39%
Audited ensemble ($w_M = 0.4$, $w_D = 0.6$)	90.76%	91.01%	85.18%

The equal-weight result indicates that the two models contain complementary information on this validation partition. However, the gain over the saved MobileNetV2 checkpoint is only 0.40 percentage points and is measured on a partition with documented lesion overlap. It should therefore be interpreted as a comparative validation observation rather than evidence of independent clinical improvement.

4.5. Compression and Host-CPU Deployment Analysis

Table 4 summarizes the principal compression and deployment measurements. Full INT8 TFLite MobileNetV2 reduced storage by 90.27% relative to the FP32 checkpoint and retained accuracy within 0.24 percentage points. It also achieved 4.51 ms batch-one latency on the benchmark host. In contrast, 50% unstructured pruning reduced accuracy by 7.47 percentage points and Melanoma recall by 13.61 percentage points without providing a dense CPU speedup. DINOv2 FP32 required 330.34 MiB on disk and showed 235.09–373.35 ms host latency. Native dynamic INT8 reduced DINOv2 storage to 87.38 MiB and latency to 173.54 ms, establishing a compact optimization baseline of 61.59% accuracy and 49.67% Melanoma recall for future quantization-aware improvement.

Table 4. Consolidated validation and host-CPU deployment measurements.

Model	Accuracy	Macro F1	Mel. Recall	Size / Latency
MobileNetV2 FP32	90.96%	91.27%	87.06%	30.21 MiB / 104.89 ms
MobileNetV2 Dynamic TFLite	90.88%	91.20%	85.62%	2.74 MiB / 27.76 ms
MobileNetV2 Full INT8 TFLite	**90.72%**	**91.03%**	**86.73%**	**2.94 MiB / 4.51 ms**
MobileNetV2 50% Pruned	83.49%	83.55%	73.45%	30.21 MiB / 101.54 ms
DINOv2 FP32	86.66%	86.82%	79.09%	330.34 MiB / 235–373 ms
DINOv2 Dynamic INT8	61.59%	57.54%	49.67%	87.38 MiB / 173.54 ms

The DINOv2 INT8 result provides a useful optimization diagnosis. Layer-wise diagnostics identified extreme activation outliers in transformer Blocks 8–11, with magnitudes up to approximately 494.72 compared with upper-percentile values near 1.3–2.7 in the corresponding diagnostic summaries. These observations identify the principal target for quantization-aware training, per-channel scaling, calibration refinement, or ViT-S distillation before deployment.

4.6. Confusion Matrix and Error Analysis

Figure 6 presents the confusion matrix of the best-performing weighted ensemble for the three target classes: BCC, Melanoma, and Nevus.The diagonal elements represent correctly classified samples, while the off-diagonal elements represent misclassifications.

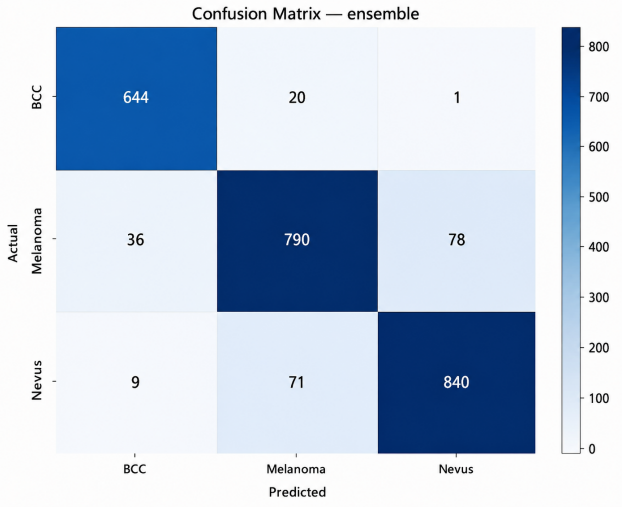


Fig. 6. Confusion matrix of the best weighted ensemble for BCC, Melanoma, and Nevus classification.

For the BCC class, 644 images were correctly classified. Twenty BCC images were predicted as Melanoma, while one image was predicted as Nevus.For the Melanoma class, 790 images were correctly classified. Thirty-six Melanoma images were predicted as BCC, while 78 images were predicted as Nevus.For the Nevus class, 840 images were correctly classified. Nine Nevus images were predicted as BCC, while 71 Nevus images were predicted as Melanoma.

The confusion matrix demonstrates that the largest classification confusion occurs between Melanoma and Nevus.

Table 5 presents the class-level precision, recall, and F1-score values derived from the confusion matrix.

BCC achieved the highest recall, indicating strong performance for this category. The Melanoma class achieved the lowest recall among the three classes, with a substantial number of Melanoma images being predicted as Nevus.

The confusion between Melanoma and Nevus represents the main source of classification error. In particular, 78 Melanoma images were predicted as Nevus, while 71 Nevus images were predicted as Melanoma.

Table 5. Class-level performance of the best weighted ensemble.

Class	Support	Precision	Recall	F1-score
BCC	665	93.47%	96.84%	95.13%
Melanoma	904	89.67%	87.39%	88.52%
Nevus	920	91.40%	91.30%	91.35%
Macro avg	2,489	91.51%	91.84%	91.67%

5. Discussion

The experimental results demonstrate that MobileNetV2 provides the strongest accuracy–efficiency trade-off among the evaluated variants. Its FP32 checkpoint achieved 90.96% validation accuracy and 87.06% Melanoma recall, while Full INT8 conversion retained 90.72% accuracy and 86.73% Melanoma recall with a substantially smaller model and lower host-CPU latency. This supports Full INT8 MobileNetV2 as a practical candidate for subsequent edge testing.

The equal-weight ensemble achieved 91.36% on the legacy validation partition, only 0.40 percentage points above the saved MobileNetV2 checkpoint. The previously reported 0.4/0.6 attribution was incorrect: that configuration achieved 90.76% and 85.18% Melanoma recall. Because ensemble weights were evaluated using the same validation predictions, the apparent gain is subject to selection bias. The lesion-overlap audit further showed a 7.79 percentage-point accuracy gap between overlapping and clean validation images for the equal-weight ensemble. These results demonstrate complementary predictions, but do not establish independent test-set improvement.

The class-level analysis shows that Melanoma remains the most important error category, particularly because Melanoma images are confused with Nevus. The 50% pruning experiment further shows that reducing parameter magnitudes without a suitable sparse execution strategy can harm this clinically relevant class. The DINOv2 INT8 experiment complements this result by identifying activation calibration as the main optimization bottleneck for transformer compression.

The framework is integrated into a Streamlit application, but the current application uses Dynamic-Range TFLite rather than the Full INT8 candidate. The reported timing measurements were obtained on an Intel Core i5-1235U host. No physical Raspberry Pi benchmark, MobileNetV2 peak-RSS measurement, statistical significance test, skin-tone fairness evaluation, or lesion-disjoint retrained test result is available in the present study. The dataset also contains only three of the eight ISIC 2019 categories, includes 1,886 images without lesion IDs, and has no external multi-centre validation. These limitations prevent clinical generalization or an unconditional Raspberry Pi deployment claim.

6. Prediction Results

Representative skin lesion images and their predicted classes are shown in Figure 7. The confidence score for each prediction is displayed along with the corresponding lesion category. The examples include BCC, Melanoma, and Nevus images.

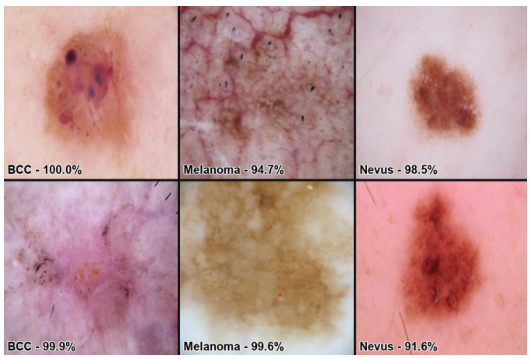


Fig. 7. Representative skin lesion images with their predicted classes and confidence scores.

7. Conclusion and Future Work

This work presented a comparative study of MobileNetV2 and DINOv2 for three-class skin lesion classification. On the legacy image-level validation partition, MobileNetV2 FP32 achieved 90.96% accuracy, while Full INT8 TFLite reduced storage to 2.94 MiB and retained 90.72% accuracy with 86.73% Melanoma recall. The equal-weight ensemble reached 91.36%, but this result was obtained on a validation partition with 54.04% lesion overlap and therefore cannot be interpreted as independent test performance. The experiments established Full INT8 MobileNetV2 as the current practical edge baseline and identified the activation-calibration strategy required to improve DINOv2 compression.

The main practical conclusion is that Full INT8 MobileNetV2 is the most promising candidate for immediate edge evaluation, while the DINOv2 INT8 result provides a measurable starting point for quantization-aware redesign and distillation. The present results do not establish clinical generalization or physical Raspberry Pi performance.

Future work will retrain both branches using the prepared zero-overlap lesion-grouped split, freeze ensemble weights and any Melanoma safety threshold on validation data, and perform one final held-out test evaluation. Further work should include paired statistical tests, confidence intervals, Raspberry Pi latency and memory measurements, attention rollout for DINOv2, skin-tone fairness analysis, calibration, and comparison with external datasets. Extending the study to all eight ISIC 2019 categories, quantization-aware training, ViT-S distillation, out-of-distribution rejection, and imbalance-aware losses may improve the reliability of a future deployment-oriented system.

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References

[1] Larxel, "Skin Lesion Images for Melanoma Classification," Kaggle Dataset, 2020. Available: https://www.kaggle.com/datasets/andrewmvd/isic-2019.

[2] Esteva, Andre, Brett Kuprel, Roberto A. Novoa, Justin Ko, Susan M. Swetter, Helen M. Blau, and Sebastian Thrun. (2017) "Dermatologist-level classification of skin cancer with deep neural networks." *Nature* **542** (7639): 115–118.

[3] Sandler, Mark, Andrew Howard, Menglong Zhu, Andrey Zhmoginov, and Liang-Chieh Chen. (2018) "MobileNetV2: Inverted residuals and linear bottlenecks." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*: 4510–4520.

[4] Howard, Andrew G., Menglong Zhu, Bo Chen, Dmitry Kalenichenko, Weijun Wang, Tobias Weyand, Marco Andreetto, and Hartwig Adam. (2017) "MobileNets: Efficient convolutional neural networks for mobile vision applications." *arXiv preprint* arXiv:1704.04861.

[5] Oquab, Maxime, Timothée Darcet, Théo Moutakanni, Huy Vo, Marc Szafraniec, Vasil Khalidov, Pierre Fernandez, et al. (2024) "DINOv2: Learning robust visual features without supervision." *Transactions on Machine Learning Research*.

[6] Caron, Mathilde, Hugo Touvron, Ishan Misra, Hervé Jégou, Julien Mairal, Piotr Bojanowski, and Armand Joulin. (2021) "Emerging properties in self-supervised vision transformers." *Proceedings of the IEEE/CVF International Conference on Computer Vision*: 9650–9660.

[7] Dosovitskiy, Alexey, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, et al. (2021) "An image is worth 16x16 words: Transformers for image recognition at scale." *International Conference on Learning Representations*.

[8] He, Kaiming, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. (2016) "Deep residual learning for image recognition." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*: 770–778.

[9] Deng, Jia, Wei Dong, Richard Socher, Li-Jia Li, Kai Li, and Li Fei-Fei. (2009) "ImageNet: A large-scale hierarchical image database." *Proceedings of the IEEE Conference on Computer Vision and Pattern Recognition*: 248–255.

[10] Codella, Noel C. F., David Gutman, M. Emre Celebi, Brian Helba, Michael A. Marchetti, Stephen W. Dusza, Aadi Kalloo, et al. (2018) "Skin lesion analysis toward melanoma detection: A challenge at the 2017 International Symposium on Biomedical Imaging (ISBI), hosted by the International Skin Imaging Collaboration (ISIC)." *IEEE 15th International Symposium on Biomedical Imaging*: 168–172.

[11] Tschandl, Philipp, Cliff Rosendahl, and Harald Kittler. (2018) "The HAM10000 dataset, a large collection of multi-source dermatoscopic images of common pigmented skin lesions." *Scientific Data* **5**: 180161.

[12] Combalia, Marc, Noel C. F. Codella, Veronica Rotemberg, Brian Helba, Veronica Vilaplana, Ofer Reiter, Cristina Carrera, et al. (2019) "BCN20000: Dermoscopic lesions in the wild." *arXiv preprint* arXiv:1908.02288.

[13] Kittler, Josef, Mohamad Hatef, Robert P. W. Duin, and Jiri Matas. (1998) "On combining classifiers." *IEEE Transactions on Pattern Analysis and Machine Intelligence* **20** (3): 226–239.

[14] Dietterich, Thomas G. (2000) "Ensemble methods in machine learning." *Multiple Classifier Systems, Lecture Notes in Computer Science* **1857**: 1–15.